

Response-Memory Closures for Learned Increments in Deep Nonlinear Gradient Flow

Fixed-width trajectory approximation on compact horizons

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Abstract

We study an autonomous approximation of the complete parameter trajectory of fixed-finite-width deep nonlinear gradient flow. The exact learned increment of each internal matrix is a time integral of a backward response against a forward response. We project both endogenous response histories onto the first P shifted-Legendre modes while retaining the initialized matrices exactly. Online Legendre projection and its triangular coefficient dynamics are inherited from polynomial-memory methods such as HiPPO; the neural construction couples the two projected histories through the reconstructed network.

For arbitrary fixed depth, width, data and initialization, an ordinary residual-activity clock gives uniform physical-trajectory error $O_T(P^{-1})$ on every prescribed finite horizon. For positive initial residual, a joint response-speed clock gives $O_T(P^{-2})$ under one additional activation derivative. Neither result assumes successful fitting, Gaussian initialization or special input geometry. The second closure evolves $2(H-1)mnP$ history scalars and a shared symmetric $P \times P$ Gram matrix; its complete moving-state scaling under the sufficient order rule is $O(Hmn\varepsilon^{-1/2} + \varepsilon^{-1})$, before common outer blocks.

This is a representation theorem for learned increments, not a net model-memory or runtime compression theorem. Every initialized internal matrix, its forward and transpose actions, and fixed prefix data remain; reported implementations have larger minimal total state and are slower than dense flow in the large panels. Existing circle and MNIST experiments show selected low-order matched-loss predictor fidelity, but they do not measure the same-time trajectory rates. We separate the fixed-width theorem from width-uniform population closure, all-time tracking and compression of the initialized operator, none of which is proved.

1 A fixed-width trajectory-approximation problem

A snapshot compressor need only reproduce the current function. A training compressor must also reproduce how the function will move. Equality of predictions does not identify the parameter-space Jacobian, the backward responses, or the optimizer metric. Consequently, two exact functional copies can acquire different velocities and their errors can accumulate after training continues.

Our target keeps more than one hidden layer, active nonlinearities and moving hidden representations. It is stronger than matching outputs or kernels: for a fixed finite network it controls every learned matrix and the readout uniformly on $[0, T]$. The approximation state is finite for every fixed width and order and does not grow with elapsed training time. Its size can still grow with width, samples and order, and it retains every initialized dense operator.

1.1 Claim levels and state classes

We use the following distinctions throughout.

Definition 1.1 (Compact-horizon global approximation). Let $z(t)$ be a reference flow and $z_P(t)$ an autonomous order- P closure. The closure is *globally accurate on compact horizons* if, for every fixed $T < \infty$,

$$\sup_{0 \leq t \leq T} d(z_P(t), z(t)) \longrightarrow 0 \quad (P \rightarrow \infty).$$

The order may depend on T . This does not mean that one fixed order works for all $t \geq 0$, nor that the constants remain bounded as $T \rightarrow \infty$.

Three kinds of autonomous state are relevant.

1. A *finite-scalar ODE* evolves finitely many real coordinates at each fixed n, m, H, P . Its dimension may depend on those parameters but not on elapsed training time or a discretization of a population.
2. A *population or field ODE* evolves finitely many functions, random variables or densities on fixed source spaces. It has finitely many fields but infinitely many scalar degrees of freedom.
3. An *operator ODE* evolves maps between population spaces. A single operator retains the row-column incidence that a finite collection of marginal populations may lose. “One operator” therefore need not mean low information.

Exact and approximate closure are separate questions in each class. So are a local velocity match and uniform trajectory control on $[0, T]$.

1.2 Relation to prior dynamics and memory models

Online polynomial memory. HiPPO formulates online compression of a cumulative signal as projection onto a finite polynomial space under a time-dependent measure [Gu et al., 2020]. Its scaled-Legendre construction satisfies a closed triangular coefficient equation of the form

$$\dot{c}(t) = -t^{-1}Ac(t) + t^{-1}Bf(t),$$

and its approximation results quantify error for a supplied signal history. Thus online rescaled Legendre moments and their triangular dilation dynamics are direct ingredients, not claims of this paper. HiPPO does not couple endogenous forward and backward training responses, reconstruct learned matrix increments, or propagate the truncation through nonlinear optimizer feedback to a complete fixed-width parameter trajectory.

Neural Tangent Hierarchy. Huang and Yau derive an exact hierarchy for finite-width training outputs and higher tangent kernels and close it at order p by freezing the top kernel [Huang and Yau, 2020]. To translate their notation, let m denote training samples and n network width here. An order- r kernel evaluated on the training set carries up to m^r indexed values, so explicit storage through order p is dominated by $O(m^p)$ before symmetry reductions; this is a direct count from the tensor arguments, not a complexity theorem stated in that work. Under its smoothness, input-nondegeneracy, even- p and initial-kernel-gap assumptions, its Theorem 2.6 applies on

$$t \leq \min \left\{ \frac{c\sqrt{\lambda n/m}}{(\log n)^C}, \frac{n^{p/[2(p+1)]}}{(\log n)^{C'}} \right\},$$

with training-output error of the form

$$\|f - \tilde{f}\|_{\ell^2(m)} \lesssim \frac{(1+t)t^{p-1}\sqrt{m}}{n^{p/2}} \min\{t, m/\lambda\},$$

and an entrywise NTK error in its Eq. (2.10). Its Eq. (2.11) evolves one-test-point tensors through order p , requiring up to $O(m^{p-1})$ additional indexed values per test point. The supplied NTH paper gives no implementation or direct experiment of this truncation. Its controlled objects are outputs and kernels in a wide/NTK-scaled regime; ours are the complete physical parameters of a fixed finite network as moment order increases.

Feature-learning population and field descriptions. Tensor Programs IV proves leading-order feature-learning limits under maximal-update parameterization at infinite width [Yang and Hu, 2021]. Its state consists of coordinate distributions and random variables; the explicit finite-step recursion includes variables indexed by earlier training steps. Its limit is taken at each fixed discrete training time, and the paper supplies an algorithm and experiments, rather than a fixed-width parameter-trajectory approximation.

Feature-learning dynamical mean-field theory represents infinite-width training by self-consistent two-time activation, gradient and response kernels [Bordelon and Pehlevan, 2022]. On a grid with M samples and N_t time points, its Table 1 reports $O(M^2 N_t^2)$ kernel memory and $O(M^3 N_t^3)$ kernel time; it provides an alternating sampling algorithm and finite-network experiments, while its derivation is not a fixed-finite-width trajectory theorem. Its finite-width extension gives perturbative $N^{-1/2}$ fluctuations and N^{-1} mean corrections around those order parameters [Bordelon and Pehlevan, 2023], not an exact closure of one finite realization.

Rigorous multilayer mean field evolves functions $w_1(t, c_1)$, $w_\ell(t, c_{\ell-1}, c_\ell)$ and $b_\ell(t, c_\ell)$ on a fixed neuronal ensemble and couples wide finite networks to this autonomous flow on compact horizons [Nguyen and Pham, 2023]. Neural Feature Flow similarly evolves feature measures and pair functions on their supports and gives a finite-width approximation under its initialization, width and step-size conditions [Fang et al., 2020]. These systems have finitely many field types but infinitely many scalar degrees of freedom; they are width-limit descriptions, not fixed-width matrix compression. Shallow Wasserstein and distributional limits provide related one- or two-layer precedents [Chizat and Bach, 2018, Sirignano and Spiliopoulos, 2019, Mei et al., 2018].

A companion finite-type population closure. The maintained companion book proves a closer autonomous-closure result for a specific two-hidden-layer nonlinear population flow [PDE Project, 2026]. It constructs increasing finite-type systems that are restartable from their saved current state and whose whole-circle predictions and declared joint observations converge uniformly on a fixed compact interval. Their state, however, contains two probability laws plus a finite matrix of action contractions, hence infinitely many scalar degrees of freedom. The result gives neither a quantitative state/error rate nor a finite-scalar approximation of one fixed network’s full parameter path.

Lazy limits, initialization theories and projection closure. NNGP, signal-propagation and dynamical-isometry results concern random initialization and its input-output statistics [Lee et al., 2018, Schoenholz et al., 2017, Pennington et al., 2017]. The NTK and wide-network linearization give infinite-width or sufficiently wide lazy-regime function dynamics in which hidden-feature motion vanishes [Jacot et al., 2018, Lee et al., 2019]. They locate a different regime rather than attempt the present closure. More broadly, Mori–Zwanzig projection writes resolved dynamics as Markovian, memory and unresolved terms [Chorin et al., 2000]; it is conceptual memory-closure precedent, not a neural trajectory theorem.

Within this reviewed, nonexhaustive source set, none combines all of the following: fixed finite width and data, deep nonlinear feature-learning gradient flow, an elapsed-time-independent finite

Table 1: Contract-level comparison. “Finite scalar” means finite at fixed network width and order; it need not be width independent.

Work	Regime and controlled object	State / direct scaling	Horizon and implementation evidence
HiPO-LegS	supplied exogenous signal history	P polynomial coefficients	online ODE; signal-approximation results
NTH	wide/NTK-scaled outputs and kernels	training tensors through $O(m^p)$	width-dependent theorem horizon; no truncated-solver experiment in supplied source
Tensor Programs IV	infinite-width feature-learning law	coordinate random variables, including past-indexed variables	each fixed discrete step count; Algorithm 1 and experiments
Feature-learning DMFT	infinite-width kernels and single-site processes	$O(M^2 N^2)$ memory, $O(M^3 N^2)$ time	two-time grid; alternating sampler and experiments
Finite-width DMFT	fluctuations of DMFT order parameters	perturbative fields/Hessian around the width limit	finite-width corrections; not one-network exact closure
Multilayer mean field / NFF	width-limit parameter or feature fields	finitely many field types, infinitely many scalars	fixed horizon; quantitative wide-network coupling
Companion observable closure	specific two-layer population predictions/observations	two probability laws plus a finite matrix; infinitely many scalars	restartable, uniform on fixed compact time; no state/error rate
This work	one fixed finite network’s complete parameter path	finite moving state in Table 2; retained dense W_0	every prescribed finite horizon; both closures implemented

autonomous state, reconstruction of learned internal increments, and uniform approximation of the complete physical trajectory on every prescribed finite interval. We make no unrestricted priority claim.

2 Dense deep gradient flow

Fix a dataset $(x_a, y_a)_{a=1}^m$, $x_a \in \mathbb{R}^d$, a width n , and $H \geq 2$ hidden layers. Set $v_a = x_a/\sqrt{d}$. For simplicity every hidden layer has width n ; activations may differ by layer:

$$\begin{aligned}
z_a^{(1)} &= W^{(1)} v_a, & h_a^{(1)} &= \phi_1(z_a^{(1)}), \\
z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi_\ell(z_a^{(\ell)}), \quad 2 \leq \ell \leq H, \\
f_a &= \frac{w^\top h_a^{(H)}}{n}, & r_a &= f_a - y_a, \quad \rho = \left(\frac{1}{m} \sum_a r_a^2 \right)^{1/2}.
\end{aligned} \tag{1}$$

The loss is $\mathcal{L} = \rho^2$. Backward responses exclude the residual:

$$\begin{aligned}
\delta_a^{(H)} &= w \odot \phi'_H(z_a^{(H)}), \\
\delta_a^{(\ell)} &= \phi'_\ell(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}.
\end{aligned} \tag{2}$$

With mobilities $(n, 1, \dots, 1, n)$, the physical gradient field F is

$$\begin{aligned}
F_1 &= -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^\top, \\
F_\ell &= -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top, \quad 2 \leq \ell \leq H, \\
F_w &= -\frac{2}{m} \sum_a r_a h_a^{(H)}.
\end{aligned} \tag{3}$$

Thus every internal matrix has the exact representation

$$W^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t r_a(s) \delta_a^{(\ell)}(s) (h_a^{(\ell-1)}(s))^\top ds. \tag{4}$$

The growing history in (4) is the object we compress. The initialized matrices $W_0^{(\ell)}$ are retained exactly.

3 Variant A: the residual-activity clock

Let p_k be shifted Legendre polynomials on $[0, 1]$, normalized by $p_k(1) = 1$ and

$$\int_0^1 p_j(x)p_k(x) dx = \frac{\mathbf{1}_{j=k}}{2k+1}.$$

They satisfy the triangular identity

$$xp'_k(x) = kp_k(x) + \sum_{j < k} (2j+1)p_j(x). \quad (5)$$

This scaled-Legendre online-memory identity is the mechanism used by HiPPO-LegS [Gu et al., 2020]; we record the normalization because it fixes the coefficients in the neural closure.

Define the activity coordinate

$$\tau(t) = 1 + \int_0^t \rho(s) ds. \quad (6)$$

At internal link ℓ and sample a , define

$$a_{\ell a} = h_a^{(\ell-1)}, \quad b_{\ell a} = \frac{r_a}{\rho} \delta_a^{(\ell)} \quad (\rho > 0).$$

In the coordinate $\xi = \tau(s)$, extend $a_{\ell a}$ constantly over the artificial prefix $[0, 1]$ and set $b_{\ell a} = 0$ there. Store only

$$H_{\ell ak} = \int_0^\tau \bar{a}_{\ell a}(\xi) p_k(\xi/\tau) d\xi, \quad U_{\ell ak} = \int_0^\tau \bar{b}_{\ell a}(\xi) p_k(\xi/\tau) d\xi, \quad 0 \leq k < P. \quad (7)$$

Differentiating and using (5) gives the closed ODE

$$\begin{aligned} \dot{H}_{\ell ak} &= \rho a_{\ell a} - \frac{\rho}{\tau} \left[k H_{\ell ak} + \sum_{j < k} (2j+1) H_{\ell aj} \right], \\ \dot{U}_{\ell ak} &= r_a \delta_a^{(\ell)} - \frac{\rho}{\tau} \left[k U_{\ell ak} + \sum_{j < k} (2j+1) U_{\ell aj} \right]. \end{aligned} \quad (8)$$

Initially $H_{\ell a0} = a_{\ell a}(0)$ and all other moments vanish. The reconstructed matrices are

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{P-1} (2k+1) U_{\ell ak} H_{\ell ak}^\top. \quad (9)$$

All responses in (8) are evaluated in the current reconstructed network. The first matrix and readout evolve by their exact blocks in (3). Equations (6)–(9) are therefore an autonomous finite ODE. They never divide by ρ ; the normalized b is only a proof coordinate.

Conversely, along any local solution the defining history integrals satisfy (8) with the same initial values. Their difference from the ODE state solves a homogeneous linear equation with zero initial value. Thus the stored variables are exactly the first P moments of the closure’s own responses; only their projected cross-product is truncated.

If Π_P denotes componentwise $L^2(0, \tau)$ projection onto polynomials of degree less than P , then (9) is exactly

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^\tau (\Pi_P \bar{b}_{\ell a})(\xi) (\Pi_P \bar{a}_{\ell a})(\xi)^\top d\xi. \quad (10)$$

Thus closure of the moment ODE is exact; truncating the two history projections is the modeling approximation.

3.1 Compact-horizon convergence

Let $\theta = (W^{(1)}, \dots, W^{(H)}, w)$ and use the Euclidean block norm, with Frobenius norms on matrices.

Theorem 3.1 (Ordinary-clock response-memory convergence). *Fix finite n, d, m, H , finite data and finite initialization. Suppose $\phi_\ell \in C_{\text{loc}}^{1,1}(\mathbb{R})$ for every layer. For every $T < \infty$ there are finite constants $C_{\text{old}}(T)$ and $P_0(T)$, independent of P , such that the order- P closure exists uniquely through T for every $P \geq P_0(T)$ and*

$$\sup_{0 \leq t \leq T} \left\| \hat{\theta}_P(t) - \theta(t) \right\| \leq \frac{C_{\text{old}}(T)}{\sqrt{P(P+1)}}. \quad (11)$$

The same rate holds for forward and backward responses on the training set, and for forward activations and predictions on any fixed bounded collection of passive inputs.

If, in addition, every ϕ_ℓ and ϕ'_ℓ is globally bounded, then the closure exists uniquely for all finite physical times for every $P \geq 1$, and (11) holds for every $P \geq 1$ after enlarging the constant. This includes $\tanh$, logistic sigmoid, arctangent and smooth bounded periodic activations. No fitting, Gaussian initialization, input separation or label compatibility assumption is required.

The word “global” in this theorem means the whole interval $[0, T]$ for each arbitrary finite T . It does not give a T -independent constant.

3.2 Proof of the ordinary-clock theorem

We give the proof because it shows exactly where history regularity enters and why feedback does not invalidate the projection estimate.

Lemma 3.2 (Legendre tail). *For $q \in H^1(0, \tau; \mathbb{R}^N)$,*

$$\|q - \Pi_P q\|_{L^2(0, \tau)}^2 \leq \frac{1}{P(P+1)} \int_0^\tau \xi(\tau - \xi) \|q'(\xi)\|^2 d\xi \leq \frac{\tau^2}{4P(P+1)} \|q'\|_{L^2(0, \tau)}^2. \quad (12)$$

Proof. On $[0, 1]$, the shifted Legendre equation is

$$-\frac{d}{dx} \{x(1-x)p'_k(x)\} = k(k+1)p_k(x).$$

Integration by parts shows that the derivatives are orthogonal in the weight $x(1-x)$ and have squared norm $k(k+1)/(2k+1)$. Bessel’s inequality applied to q' in this weighted space gives

$$\sum_{k \geq 1} k(k+1) \frac{\|c_k\|^2}{2k+1} \leq \int_0^1 x(1-x) \|q'(x)\|^2 dx,$$

where $c_k = (2k+1) \int_0^1 q(x)p_k(x) dx$. Parseval, polynomial density and $k(k+1) \geq P(P+1)$ for $k \geq P$ give the first bound on $[0, 1]$. Rescaling and $\xi(\tau - \xi) \leq \tau^2/4$ give (12); summing componentwise proves the vector statement. $\square$

Step 1: dense existence and a common compact region. The mobility matrix $\mathcal{D}$ has largest eigenvalue n , and $F = -\mathcal{D}\nabla\mathcal{L}$. Hence

$$\dot{\mathcal{L}} = -\left\| \mathcal{D}^{-1/2} \dot{\theta} \right\|^2, \quad \int_0^t \left\| \dot{\theta}(s) \right\|^2 ds \leq n\rho(0)^2. \quad (13)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nt}\rho(0)$. A finite maximal endpoint would have a finite limiting state, from which local existence extends the solution. Dense flow is therefore global.

Fix T and let

$$R = \|\theta(0)\| + \sqrt{nT}\rho(0) + 1, \quad \mathcal{B} = \{\vartheta : \|\vartheta\| \leq R\}.$$

The dense path remains distance at least one from $\partial\mathcal{B}$. Stop the closure at its first exit from $\mathcal{B}$. The rest of the proof uses constants on this explicit ball and will rule out that exit.

For a concrete choice of constants set $X = \max_a \|v_a\|$ and $\alpha_0 = X$. Recursively choose

$$\begin{aligned} s_\ell &= \sup_{|z| \leq R\alpha_{\ell-1}} |\phi'_\ell(z)|, & \alpha_\ell &= \sqrt{n} \sup_{|z| \leq R\alpha_{\ell-1}} |\phi_\ell(z)|, \\ \beta_H &= Rs_H, & \beta_\ell &= s_\ell R\beta_{\ell+1} \quad (\ell < H). \end{aligned} \quad (14)$$

Then $\|h_a^{(\ell)}\| \leq \alpha_\ell$ and $\|\delta_a^{(\ell)}\| \leq \beta_\ell$ on $\mathcal{B}$. Put

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2 \right)^{1/2}, & q &= R\alpha_H/n + Y, & S &= Tq, & A &= 1 + S, \\ v_1 &= 2\beta_1 X, & v_\ell &= 2\beta_\ell \alpha_{\ell-1}/n, & v_w &= 2\alpha_H, \\ V &= \left(v_1^2 + \sum_{\ell=2}^H v_\ell^2 + v_w^2 \right)^{1/2}. \end{aligned} \quad (15)$$

On the stopped interval, $\rho \leq q$, $\tau \leq A$ and $\|F\| \leq V\rho$. Since F is locally Lipschitz, it has some finite Lipschitz constant K on $\mathcal{B}$.

Step 2: an exact projection-error energy. For any one forward or backward history q let $D_q(t) = \|q - \Pi_P q\|_{L^2(0,\tau(t))}^2$ and let q^* be the polynomial projection evaluated at the current endpoint. Differentiating the raw energy and the projected energy, using (8), cancels every basis-dilation term and gives

$$\dot{D}_q = \rho \|q - q^*\|^2, \quad D_q(t) = \int_0^t \rho(s) \|q(s) - q^*(s)\|^2 ds. \quad (16)$$

The backward-prefix jump is harmless: only square integrability and growing integrals are used.

Differentiate (9). The physical closure equation is exactly

$$\dot{\hat{\theta}}_P = F(\hat{\theta}_P) + E, \quad E_1 = E_w = 0, \quad (17)$$

with internal defects

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_{\ell a} - b_{\ell a}^*)(a_{\ell a} - a_{\ell a}^*)^\top. \quad (18)$$

The two mixed projection terms vanish by orthogonality. Consequently

$$\int_0^t \|E_\ell(s)\|_F ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b,\ell a}(t) D_{a,\ell a}(t)}. \quad (19)$$

This is the bridge from a same-history approximation to a trajectory theorem. Define also the proof-only accumulator

$$W_\ell^{\text{acc}}(t) = W_{\ell,0} - \frac{2}{nm} \sum_a \int_0^t \hat{r}_a \hat{\delta}_a^{(\ell)} (\hat{h}_a^{(\ell-1)})^\top ds, \quad R_\ell = \widehat{W}^{(\ell)} - W_\ell^{\text{acc}}.$$

Then $R_\ell(0) = 0$ and $\dot{R}_\ell = E_\ell$, so (19) controls accumulated absolute velocity error, not only a signed endpoint integral.

Step 3: forward histories have a P -independent derivative bound. Before changing variables, note that a nonstationary ordinary-clock solution cannot first reach $\rho = 0$ at a finite regular state. Every raw velocity vanishes there, and local uniqueness applied backward from that equilibrium would force the preceding path to be stationary. Hence $\xi = \tau(t)$ is a valid coordinate on every nonstationary compact segment; the raw algorithm itself remains defined at $\rho = 0$ without division.

Use primes for derivatives in $\xi = \tau(t)$ and define

$$Z_\ell(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \left\| (h_a^{(\ell)})'(\xi) \right\|^2 d\xi.$$

The first layer satisfies $\left\| (h_a^{(1)})' \right\| \leq s_1 v_1 X$. For later layers, the chain rule gives

$$(h_a^{(\ell)})' = \phi_\ell'(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + \widehat{W}^{(\ell)}(h_a^{(\ell-1)})' \right].$$

Endpoint evaluation of a degree- $(P-1)$ polynomial has operator norm $P/\sqrt{\tau}$ from $L^2(0, \tau)$, because $\sum_{k < P} (2k+1) = P^2$. Projection contraction and $m^{-1} \sum_a \|b_{\ell a}\|^2 \leq \beta_\ell^2$ imply

$$\left(m^{-1} \sum_a \|b_{\ell a}^*\|^2 \right)^{1/2} \leq P\beta_\ell, \quad \left(m^{-1} \sum_a \|b_{\ell a} - b_{\ell a}^*\|^2 \right)^{1/2} \leq (P+1)\beta_\ell.$$

Combining this with Lemma 3.2, (16) and (18) yields

$$\int_0^t \rho \|E_\ell/\rho\|_F^2 ds \leq \frac{2\beta_\ell^2 A^2}{n^2} Z_{\ell-1}(t). \quad (20)$$

The endpoint factor $P+1$ cancels the P^{-1} projection factor. This is the step that prevents a loss of convergence order with depth.

Define finite, P -independent bounds recursively by

$$\begin{aligned} \overline{Z}_1 &= S(s_1 v_1 X)^2, \\ \overline{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 \alpha_{\ell-1}^2 + \left(R^2 + \frac{2\beta_\ell^2 A^2 \alpha_{\ell-1}^2}{n^2} \right) \overline{Z}_{\ell-1} \right]. \end{aligned} \quad (21)$$

Integrating (3.2), using $\|x+y+z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$ and (20), proves inductively that $Z_\ell \leq \overline{Z}_\ell$.

Step 4: accumulated defect and feedback. The zero backward prefix gives $m^{-1} \sum_a D_{b, \ell a} \leq S\beta_\ell^2$. By Lemma 3.2,

$$\frac{1}{m} \sum_a D_{a, \ell a} \leq \frac{A^2 \overline{Z}_{\ell-1}}{4P(P+1)}.$$

Summing (19) over links gives

$$\int_0^t \|E(s)\| ds \leq \frac{B_{\text{old}}}{\sqrt{P(P+1)}}, \quad B_{\text{old}} = \frac{A}{n} \sum_{\ell=2}^H \beta_\ell \sqrt{S \overline{Z}_{\ell-1}}. \quad (22)$$

Subtracting dense flow from the perturbed flow and applying the Lipschitz bound on F gives

$$e(t) \leq \frac{B_{\text{old}}}{\sqrt{P(P+1)}} + K \int_0^t e(s) \, ds.$$

Gronwall's inequality therefore yields

$$\sup_{t \leq T} e(t) \leq \frac{B_{\text{old}} e^{KT}}{\sqrt{P(P+1)}}. \quad (23)$$

Choose P_0 so the right side is less than $1/2$. The closure then cannot reach $\partial\mathcal{B}$. Its moment representations, $1 \leq \tau \leq A$ and the response bounds place its whole finite state in a compact subset of the locally Lipschitz domain. A finite maximal endpoint would therefore extend, so the closure exists through T . Lipschitz dependence of network responses on θ gives the final prediction statement.

For completeness, suppose every activation and first derivative is globally bounded. Set

$$\alpha_\ell = \sqrt{n} \|\phi_\ell\|_\infty, \quad q_{\text{quad}} s_\ell = \|\phi'_\ell\|_\infty, \quad q_{\text{quad}} B_w = \sqrt{\|w(0)\|^2 + nY^2 T}.$$

The unchanged readout equation obeys

$$\frac{d}{dt} \|w\|^2 = nY^2 - 4n \frac{1}{m} \sum_a (f_a - y_a/2)^2 \leq nY^2,$$

for both dense and closure flows. Therefore $\|w\| \leq B_w$, $\rho \leq q := B_w \alpha_H / n + Y$, $S = Tq$ and $\tau \leq A := 1 + S$. Start with $\beta_H = B_w s_H$ and proceed downward. For $\ell = H, \dots, 2$, define

$$D_\ell = \left\| W_0^{(\ell)} \right\|_F + \frac{2A\beta_\ell \alpha_{\ell-1}}{n}, \quad \beta_{\ell-1} = s_{\ell-1} D_\ell \beta_\ell.$$

Projection contraction in (10) gives $\left\| \widehat{W}^{(\ell)} \right\|_F \leq D_\ell$; the backward recurrence then gives $\|\delta^{(\ell-1)}\| \leq \beta_{\ell-1}$. There is no circularity because the descent starts from the bounded readout. Finally

$$\left\| W^{(1)}(t) \right\|_F \leq \left\| W^{(1)}(0) \right\|_F + 2SX\beta_1.$$

These P -independent physical bounds, the moment integral representations and $1 \leq \tau \leq A$ put the order- P state in a compact set for every fixed P . Hence the closure continues through T for every $P \geq 1$. Repeating Steps 2–4 with these global constants gives (23) for every P . This completes the proof.

4 Variant B: a joint response-speed clock

The activity clock controls the first-layer forward history particularly cleanly, but it does not make every backward history smooth. A more symmetric coordinate parametrizes the entire encoded response path. This construction is defined for $\rho(0) > 0$. If $\rho(0) = 0$, dense gradient flow is already the constant physical trajectory; that separate trivial branch needs no normalized response or response-clock state. For positive initial residual, concatenate

$$\Psi(\theta) = \text{concat}_{\ell=2}^H \text{concat}_{a=1}^m \begin{bmatrix} a_{\ell a} \\ b_{\ell a} \end{bmatrix}, \quad \dot{L} = g = \rho + \left\| \dot{\Psi} \right\|_2, \quad L(0) = 1. \quad (24)$$

Then $\|\mathrm{d}\Psi/\mathrm{d}L\| \leq 1$. The clock changes where history is placed, but the learning integral still carries residual mass. At time s , place a point at $\xi = L(s)$ and assign mass $\rho(s) \mathrm{d}s$. Give $[0, 1]$ Lebesgue mass and matching constant prefixes. Thus

$$\int q \mathrm{d}\mu_t = \int_0^1 q(\xi) \mathrm{d}\xi + \int_0^t q(L(s))\rho(s) \mathrm{d}s, \quad \mathrm{d}\mu_t \leq \mathrm{d}\xi.$$

The mass is $A(t) = 1 + \int_0^t \rho \mathrm{d}s$, generally different from $L(t)$.

Let $p = (p_0, \dots, p_{P-1})^\top$, $e = (1, \dots, 1)^\top$, and let $\mathsf{T}_{kk} = k$, $\mathsf{T}_{kj} = 2j + 1$ for $j < k$. Store one shared Gram matrix and link/sample moments

$$H_{\ell a} = \int \bar{a}_{\ell a} p^\top \mathrm{d}\mu, \quad U_{\ell a} = \int \bar{b}_{\ell a} p^\top \mathrm{d}\mu, \quad G = \int p p^\top \mathrm{d}\mu,$$

where p is evaluated at $\xi/L(t)$. Put $C_{\ell a} = b_{\ell a}(0)a_{\ell a}(0)^\top$. The reconstruction and exact moment equations are

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a (U_{\ell a} G^{-1} H_{\ell a}^\top - C_{\ell a}), \quad (25)$$

$$\begin{aligned} \dot{H}_{\ell a} &= \rho a_{\ell a} e^\top - (g/L) H_{\ell a} \mathsf{T}^\top, & \dot{U}_{\ell a} &= r_a \delta_a^{(\ell)} e^\top - (g/L) U_{\ell a} \mathsf{T}^\top, \\ \dot{G} &= \rho e e^\top - (g/L) (\mathsf{T} G + G \mathsf{T}^\top). \end{aligned} \quad (26)$$

Initially $H_{\ell a} = [a_{\ell a}(0), 0, \dots]$, $U_{\ell a} = [b_{\ell a}(0), 0, \dots]$ and $G = \text{diag}(1, (1/3), \dots, (1/(2P-1)))$. The prefix correction makes $\widehat{W}^{(\ell)}(0) = W_0^{(\ell)}$.

Solve $Gq = e$ and set $a_{\ell a}^* = H_{\ell a} q$ and $b_{\ell a}^* = U_{\ell a} q$. Product differentiation cancels every g/L term and gives the physical defect

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_{\ell a} - b_{\ell a}^*)(a_{\ell a} - a_{\ell a}^*)^\top. \quad (27)$$

This has the same product-of-errors form as Variant A, but now both factors belong to one path with unit speed in L .

Theorem 4.1 (Response-clock convergence). *Under the finite setup of Theorem 3.1, assume $\phi_1 \in C_{\text{loc}}^{1,1}$ and $\phi_\ell \in C_{\text{loc}}^{2,1}$ for $\ell \geq 2$. If $\rho(0) > 0$, then for every finite T there are $P_0(T)$, Λ_T and C_T , independent of P , such that for $P \geq P_0(T)$ the ODE (24)–(26) has a unique regular solution on $[0, T]$, $L_P(t) \leq \Lambda_T$, and*

$$\int_0^T \|E(t)\| \mathrm{d}t \leq \frac{C_T}{P(P+1)}, \quad \sup_{t \leq T} \|\widehat{\theta}_P(t) - \theta(t)\| \leq \frac{C_T e^{K_T T}}{P(P+1)}. \quad (28)$$

The conclusion is $O_T(P^{-2})$ at fixed width, depth and data. It is not an all-small-order or uniform-all-time statement. The zero-loss physical branch described above is separate from this normalized-response ODE.

Proof. Assume $\rho_0 > 0$ as stated. Use the compact physical ball $\mathcal{B}$. Retain the constants $\alpha_\ell, \beta_\ell, q, S, \mathbf{A}, V, K$ from the proof of Theorem 3.1. On this ball the output map has a finite Lipschitz constant c_f : prediction and residual RMS change by at most $c_f \|\theta - \theta'\|$. Along dense flow,

$$|\dot{\rho}_{\text{dense}}| \leq c_f \|F(\theta)\| \leq c_f V \rho_{\text{dense}}.$$

Consequently

$$\rho_{\text{dense}}(t) \geq \mu := \rho_0 e^{-c_f V T} > 0, \quad 0 \leq t \leq T. \quad (29)$$

Consider the maximal regular closure interval stopped before either its physical state leaves $\mathcal{B}$ or its residual reaches $\mu/2$. We first derive a clock bound on this stopped interval without assuming small projection error. For any inserted vector history f , let

$$D_f = \|f - \Pi_t f\|_{L^2(\mu_t)}^2 = \int \|f\|^2 d\mu_t - \text{tr}(B_f G^{-1} B_f^\top).$$

Differentiating $G^{-1}G = I$ gives

$$\frac{d}{dt} G^{-1} = -\rho G^{-1} e e^\top G^{-1} + (g/L)(G^{-1} \mathbb{T} + \mathbb{T}^\top G^{-1}).$$

Together with (26), this cancels every dilation term and yields

$$\dot{D}_f = \rho \|f - f^*\|^2, \quad D_f(t) = \int_0^t \rho \|f - f^*\|^2 ds. \quad (30)$$

The initial error is zero because every prefix is constant. Thus (19) holds with the weighted errors in (30).

Put

$$Q = m \sum_{\ell=2}^H (\alpha_{\ell-1}^2 + \beta_\ell^2), \quad C_0 = \frac{AQ}{nm}.$$

The total mass of μ_t is at most $A = 1 + S$. Projection error is no larger than raw history energy, including the matching prefixes. Applying $2\sqrt{uv} \leq u + v$ in the defect estimate gives the order-independent bounds

$$\int_0^t \|E\| ds \leq C_0, \quad \int_0^t \|\dot{\hat{\theta}}\| ds \leq VS + C_0. \quad (31)$$

The second inequality uses $\dot{\hat{\theta}} = F(\hat{\theta}) + E$ and $\int_0^t \|F\| ds \leq V \int_0^t \rho ds \leq VS$. This total-variation estimate is stronger than bounded physical position and is obtained before any clock bound.

It remains to control the derivative of the normalized response map. Let $q_r = r/\rho \in \mathbb{R}^m$. Then

$$\|q_r\|^2 = m, \quad Dq_r[v] = \left(I - \frac{q_r q_r^\top}{m} \right) \frac{Dr[v]}{\rho}. \quad (32)$$

The matrix in parentheses is an orthogonal projector and $\rho \geq \mu/2$ on the stopped interval. Differentiating the forward and backward recurrences uses only bounded physical weights and bounded first two activation derivatives on compact preactivation intervals; the first layer uses only ϕ'_1 . Hence there is a finite, P -independent constant J such that

$$\|D\Psi(\vartheta)\|_{\text{op}} \leq J$$

throughout the stopped tube. The stated local smoothness also makes this derivative locally Lipschitz. Since $\dot{\Psi} = D\Psi(\hat{\theta})\dot{\hat{\theta}}$, (31) now gives

$$L_P(t) = 1 + \int_0^t (\rho + \|\dot{\Psi}\|) ds \leq A + J(VS + C_0) =: \Lambda_T. \quad (33)$$

No clock cap has been assumed or inserted.

We can now use the sharper approximation estimate. The entire concatenated history $\bar{\Psi}(\xi)$ is constant on the prefix, continuous at its join and, on the physical history,

$$\|\partial_\xi \bar{\Psi}\| = \frac{\|\dot{\Psi}\|}{\rho + \|\dot{\Psi}\|} \leq 1.$$

Therefore $\int_0^{L_P} \|\bar{\Psi}'\|^2 d\xi \leq L_P - 1$. The weighted projection is a best approximation in $L^2(\mu_t)$, and $d\mu_t \leq d\xi$, so comparison with the ordinary Legendre projection and Lemma 3.2 give the joint bound

$$\sum_{\ell,a} (D_{a,\ell a} + D_{b,\ell a}) \leq \frac{L_P^2(L_P - 1)}{4P(P + 1)} \leq \frac{\Lambda_T^2(\Lambda_T - 1)}{4P(P + 1)}. \quad (34)$$

This is one estimate for the concatenated path; it does not introduce a separate dimension factor per unit or layer.

Combining (19), (33), (34) and $2\sqrt{uv} \leq u + v$ yields

$$\int_0^t \|E\| ds \leq \frac{\Gamma_B}{P(P + 1)}, \quad \Gamma_B = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}. \quad (35)$$

Subtracting dense flow and applying the Lipschitz constant K and Gronwall gives

$$\|\hat{\theta}_P(t) - \theta(t)\| \leq \frac{\Gamma_B e^{KT}}{P(P + 1)}. \quad (36)$$

Choose $P_0(T)$ so that, for every $P \geq P_0(T)$,

$$\frac{\Gamma_B e^{KT}}{P(P + 1)} \leq \frac{1}{2}, \quad c_f \frac{\Gamma_B e^{KT}}{P(P + 1)} \leq \frac{\mu}{4}. \quad (37)$$

The first inequality rules out physical exit because the dense path stays distance one from $\partial\mathcal{B}$. The second and (29) give $\hat{\rho} \geq 3\mu/4 > \mu/2$, ruling out residual exit.

Finally the matching prefix supplies, at every fixed P ,

$$G(t) \succeq \int_0^1 p(\xi/L_P(t)) p(\xi/L_P(t))^\top d\xi. \quad (38)$$

The right side is positive definite: a zero quadratic form would make a nonzero polynomial vanish on an interval. Its entries vary continuously with L_P , so on $1 \leq L_P \leq \Lambda_T$ its least eigenvalue has a positive minimum $\gamma_{P,T}$. This minimum need not be uniform in P . The moment integral representations, finite history mass, bounded responses and bounded fixed-order polynomials then bound every raw coordinate. Together with the physical, residual, clock and Gram bounds, the full state lies in a compact subset of its locally Lipschitz domain. A finite maximal endpoint therefore extends, proving continuation through T and completing the theorem with $C_T = \Gamma_B$ and $K_T = K$. $\square$

Table 2: Exact scalar state counts. The Variant-B theory minimum packs the symmetric Gram; the implementation stores all P^2 Gram entries. Fixed prefix data are shown in factorized form rather than as $n \times n$ outer products.

	Dense gradient flow	Variant A	Variant B
History factors only	—	$2(H-1)mnP$	$2(H-1)mnP$
Common moving outer blocks	$nd+n$	$nd+n$	$nd+n$
Other moving coordinates	—	one clock	$P(P+1)/2$ Gram entries and one clock
Complete moving state	$nd+n+(H-1)n^2$	$nd+n+2(H-1)mnP+1$	$nd+n+2(H-1)mnP+P(P+1)/2+1$
Fixed internal source	—	$(H-1)n^2$ for W_0	$(H-1)n^2$ for W_0 , plus $2(H-1)mn$ prefix factors
Learned rank/link	up to n	$\leq mP$	$\leq m(P+1)$

4.1 The new clock is fully implementable

Equation (24) appears implicit only because $\dot{\Psi}$ is written compactly. No nonlinear solve for g is needed. First compute $q = G^{-1}e$, endpoint fits a^* , b^* , and internal velocities

$$V_\ell = \dot{\widehat{W}}^{(\ell)} = -\frac{2\rho}{nm} \sum_a \left[b_{\ell a} (a_{\ell a}^*)^\top + b_{\ell a}^* (a_{\ell a} - a_{\ell a}^*)^\top \right]. \quad (39)$$

These expressions contain no g . Compute $V_1 = F_1$ and $V_w = F_w$, then a directional forward pass

$$\dot{z}_a^{(1)} = V_1 v_a, \quad \dot{z}_a^{(\ell)} = V_\ell h_a^{(\ell-1)} + \widehat{W}^{(\ell)} \dot{h}_a^{(\ell-1)}, \quad \dot{h}_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot \dot{z}_a^{(\ell)}.$$

This gives $\dot{r}$ and $\dot{\rho}$. A reverse directional pass differentiates the backward recurrence, using ϕ''_ℓ for $\ell \geq 2$, and then

$$\dot{b}_{\ell a} = \frac{\dot{r}_a \delta_a^{(\ell)} + r_a \dot{\delta}_a^{(\ell)}}{\rho} - b_{\ell a} \frac{\dot{\rho}}{\rho}.$$

Now evaluate $g = \rho + \|\dot{\Psi}\|$ and finally (26). Matrix and transpose actions use the same reconstructed operator. One factors G and solves linear systems; an explicit inverse is unnecessary.

The response clock is more regular in approximation coordinates, but it is not automatically better at low order. At $P = 1$ the polynomial space is constant, so changing the coordinate alone leaves the physical constant-mode closure unchanged. Higher orders pay for Gram solves and derivative passes, and the theoretical constants can be large.

5 What is compressed?

Let H be hidden depth, n width, m training samples and P moment order. There are three different counts: history factors representing the learned increment, the complete autonomous moving state, and total stored state after fixed source data are included. They must not be interchanged.

The learned action is matrix-free:

$$\widehat{W}^{(\ell)} v = W_0^{(\ell)} v - \frac{2}{nm} \sum_a U_{\ell a} G^{-1} H_{\ell a}^\top v \quad (\text{with the prefix correction in Variant B}),$$

and the transpose uses the same factors in reverse order. A learned-correction action costs $O(nmP)$ per vector. A *total* closure action also applies W_0 or $W_0^\top$ and therefore costs $O(n^2 + nmP)$ per vector. Across a batch of m vectors, each link and direction costs $O(mn^2 + nm^2P)$. Variant B additionally performs one shared $O(P^3)$ Gram factorization, $O((H-1)nmP^2)$ factor solves per

right-hand-side evaluation, Gram and moment dilation, and directional forward/reverse passes. The implementation solves linear systems and does not form G^{-1} .

The histories alone are smaller than the dense internal moving matrices exactly when

$$2(H-1)mnP < (H-1)n^2 \iff 2mP < n. \quad (40)$$

Including common outer blocks, the complete Variant-A moving state is smaller than dense exactly when

$$2(H-1)mnP + 1 < (H-1)n^2,$$

and the symmetry-packed Variant-B moving state is smaller exactly when

$$2(H-1)mnP + \frac{P(P+1)}{2} + 1 < (H-1)n^2. \quad (41)$$

The convergence theorems do not show that an order meeting a requested error and $P \geq P_0(T)$ also meets these inequalities. Indeed, at fixed n, m, H , increasing P eventually crosses the dense count.

For restartable total storage, Variant A adds $(H-1)n^2$ fixed initialized entries to its complete moving state. Variant B also adds the factorized matching prefixes, minimally $2(H-1)mn$ scalars if saved, and the current implementation stores P^2 rather than $P(P+1)/2$ Gram entries. Either total is strictly larger than a minimal dense implementation storing only its current matrices and outer blocks. Storage is independent of elapsed training steps, but the current result is compression of the *learned-increment representation*, not net model-memory reduction.

Ignoring constants and the order threshold, Theorem 3.1 gives the sufficient scaling $P = O(\varepsilon^{-1})$ for trajectory error ε , while Theorem 4.1 gives $P = O(\varepsilon^{-1/2})$. Variant A's history-factor scaling is

$$O(Hmn\varepsilon^{-1}).$$

Variant B's history factors scale as $O(Hmn\varepsilon^{-1/2})$, while its complete moving internal state includes the Gram and scales as

$$O(Hmn\varepsilon^{-1/2} + \varepsilon^{-1}). \quad (42)$$

At fixed n, m, H the Gram term eventually dominates as $\varepsilon \downarrow 0$. These are sufficient approximation scalings, not practical speed laws: constants and order thresholds depend on the task, width, depth and horizon, and no useful-order crossover is proved.

5.1 Population interpretation and its precise limit

For fixed P , the neuron-indexed vectors can be viewed as particles sampling population fields indexed by the initialization source. This makes the closure a natural Monte Carlo implementation of a hybrid population/operator system: the response fields move, while the initialized random operator and its adjoint remain exact. The theorems above, however, keep n fixed and their constants may depend on n .

There is still a useful diagonal statement. Suppose an independent theorem establishes that dense width- n flows θ_n converge on $[0, T]$ to a population object Θ in a metric d , and suppose the observables entering d are controlled by Theorem 3.1 or Theorem 4.1. Choose any $n_j \rightarrow \infty$, then choose P_j large enough that the fixed-width closure error is at most $1/j$. The triangle inequality gives

$$d(\hat{\theta}_{n_j, P_j}, \Theta) \leq d(\hat{\theta}_{n_j, P_j}, \theta_{n_j}) + d(\theta_{n_j}, \Theta) \rightarrow 0.$$

This constructs a sequential approximation to the population flow. It does not prove that a fixed P works uniformly in width, that the W_0 action has a finite-population representation, or that the

two limits commute. The multilayer mean-field and DMFT results compared in the introduction concern different regimes, state spaces and observables [Nguyen and Pham, 2023, Bordelon and Pehlevan, 2022]; matching them is a separate theorem.

6 Evolving response moments and a rank-matched control

Response memory keeps a low-rank learned correction at every instant, but its atoms are moments of the *current closure trajectory*. The loop is

$$\text{moments} \longrightarrow \widehat{W} \longrightarrow \text{current forward/backward responses} \longrightarrow \text{moments}.$$

The basis in history coordinate is fixed; the neuron vectors multiplying that basis coevolve nonlinearly. This is why the method can represent substantial feature learning without being a frozen tangent model.

A frozen rank-matched control retained the same W_0 and trained $W = W_0 + AB$ by Euclidean gradient flow of both factors. It used two prescribed factor seeds and ranks mP , matching the two factor arrays to the $2nmP$ response-history coordinate budget. With $A(0) = 0$ and independent $B_{ij}(0) \sim N(0, 1/r)$, it matches the initialized network exactly, but its initial middle-matrix velocity matches dense flow only in expectation over B , not for either realized seed. Across five tasks and $P = 1, 3, 7$, response moments had lower matched-loss whole-circle error in all 29 numerically valid factor-seed comparisons. One hard-outlier $P = 1$ factor cell produced two capped trajectories that failed to reach the common loss threshold and was excluded rather than scored: 58 of the 60 planned factor trajectories fitted. On hard outliers at rank 56, moment error was 0.004663 versus 0.516210 and 0.251026 for the two factor seeds. The comparison concerns this factor metric and matched rank, not identical complete state or compute; it does not establish superiority over all trainable low-rank methods.

7 Empirical evidence

The experiments use finite networks and numerical time integration. Circle scores compare predictors on 8192 passive angles; MNIST scores use held-out images. Most comparisons match each model at its own first training-MSE 0.001 crossing. They therefore test agreement of fitted functions, while the theorems control same-time trajectories. The panels are selected finite-instance studies, generally with one initialization, and do not estimate the asymptotic theorem constants, same-time physical-parameter error, or an order-rate slope. Solver refinements are empirical sensitivity checks, not certified continuous-flow error bounds. The three figures below use the ordinary activity clock.

Variant B has also been implemented with batched Gram solves and directional response derivatives. In a matched width-2048 comparison on three circle tasks with tanh depth two and GELU depth three, all 42 main-panel models and seven models in a predeclared step-halving check reached training RMS at most 0.05. New-clock $P = 3$ predictor discrepancy was below 0.05 in five of six principal groups. The exception, GELU depth-three alternating quadrant, was 1.24526 at maximum step $1/64$ and 0.11200 at $1/128$. Increasing order did not remove this finite-step sensitivity: new-clock $P = 5$ was 0.99568 and 0.10313 at the two steps, while old-clock $P = 5$ was 0.01279 and 0.02150. The largest main-panel Gram condition number was approximately 330.9. Among 21 matched old/new rows, new-clock runtime was 1.79–3.99 times old-clock runtime (median 2.16).

In a broader 82-case response-clock sweep, dense, $P = 1$, $P = 2$ and $P = 3$ reached the strict training target in 79, 70, 67 and 66 cases. On those fitted pairs the median predictor discrepancies

Own-endpoint functions on the circle
Width 2048; each model at its own training-MSE 0.001 endpoint; closure retains dense W0

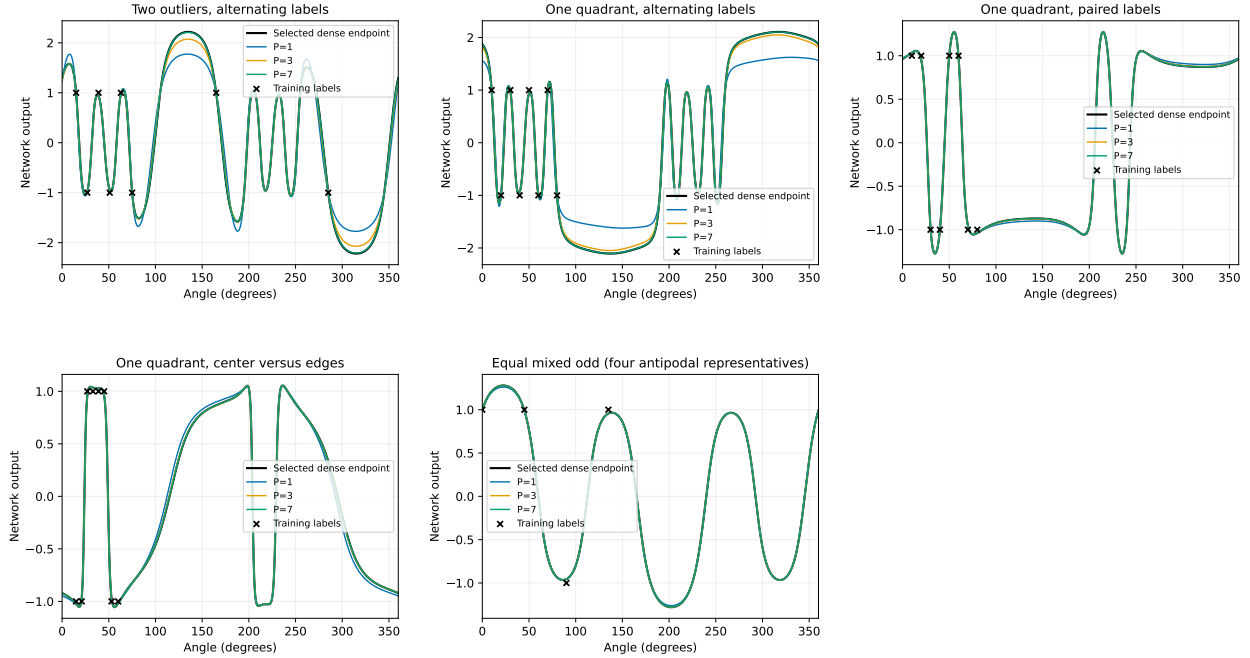


Figure 1: Two-hidden-layer tanh networks of width 2048. The black curve is the dense fitted function on the whole circle; colored curves are ordinary-clock closures at their separate first training-MSE 0.001 crossings. Training labels are marked by crosses. These are selected one-initialization matched-loss comparisons, not same-time trajectory errors.

were 0.042725, 0.018895, 0.006795, while the maxima were 3.661275, 3.240721, 1.362737; 17 cases contained at least one miss. Several activations in that sweep lie outside the smooth theorem. These runs establish implementability and some useful low-order cases. They do not empirically certify the $O(P^{-2})$ rate, continuous-flow accuracy in the sensitive case, or uniform advantage over Variant A.

Across the five tasks in Figure 1, the measured RMS errors are:

Task	$P = 1$	$P = 3$	$P = 7$
Two outliers, alternating	0.2363	0.0730	0.00466
Quadrant, alternating	0.3414	0.0457	0.00165
Quadrant, paired	0.0193	0.00282	≈ 0.00012
Quadrant, center/edges	0.0456	0.00487	0.00130
Equally spaced, mixed odd	0.0124	0.000520	≈ 0.00001

These five tasks were selected as stress cases and use one shared initialization. All displayed values use the frozen historical dense references; fresh dense refinements preserve the order trend. The paired-label and mixed-odd $P = 7$ errors are resolved only at roughly 10^{-4} and 10^{-5} scales, respectively, rather than at every printed digit. Training-point agreement alone would not force the observed whole-circle match, but this panel does not test the theorem's same-time physical trajectory.

Functions on the circle at matched training loss

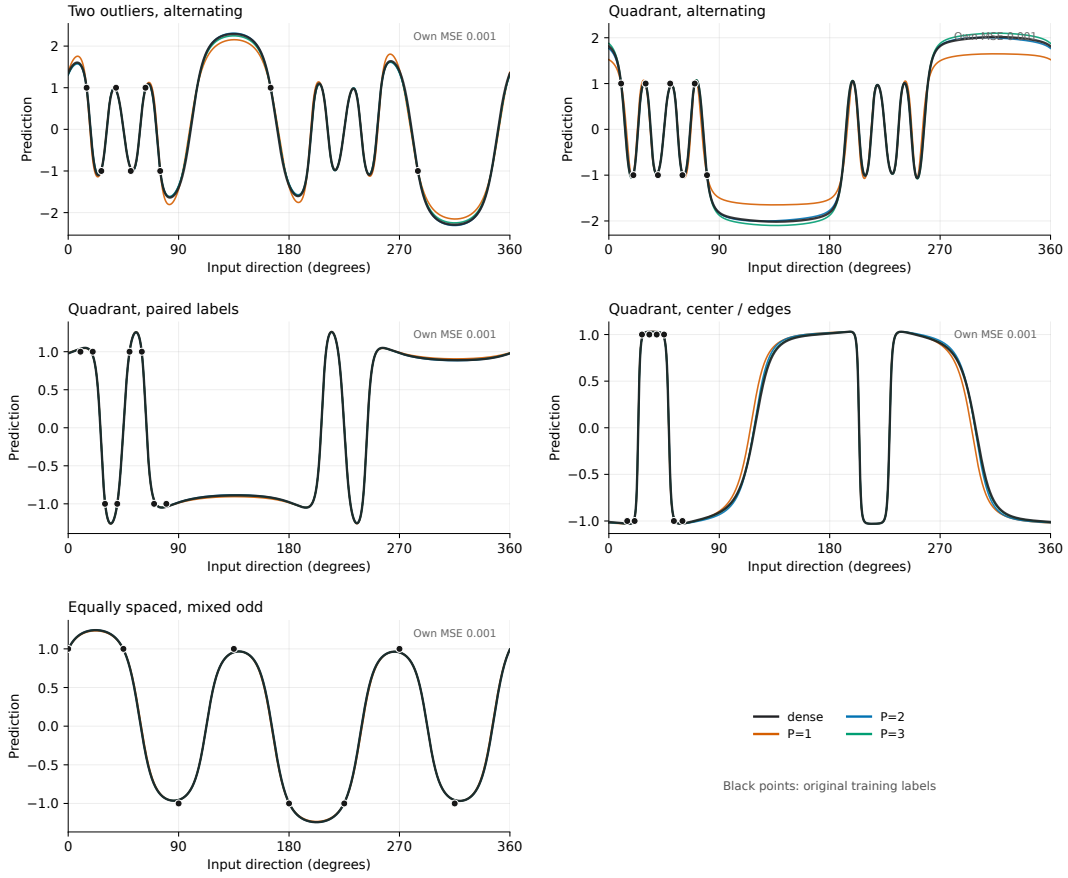
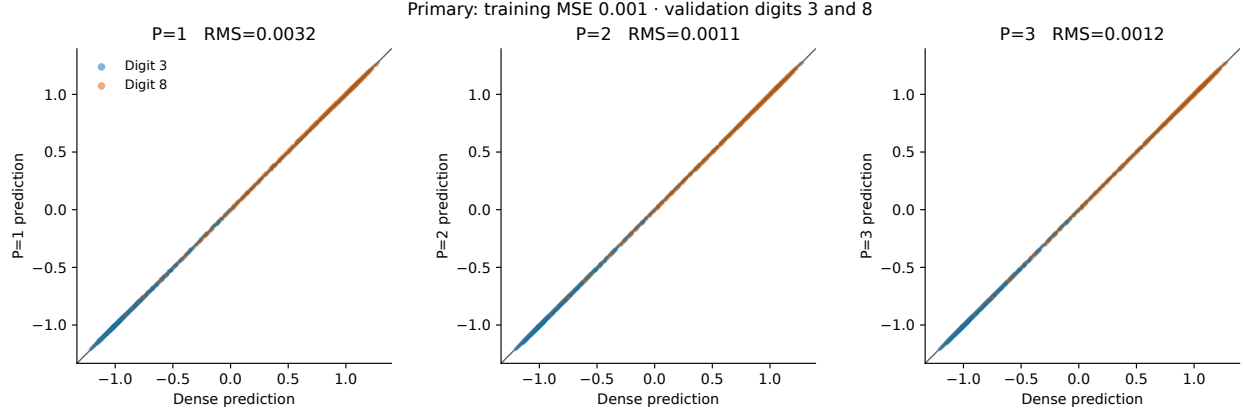
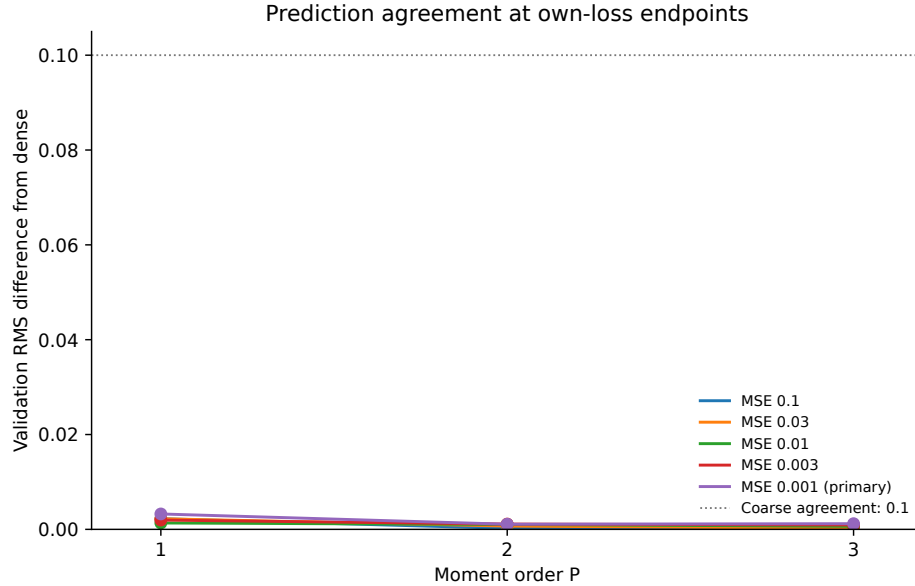


Figure 2: Three-hidden-layer tanh networks of width 4096, compared at matched training loss. Across the five tasks, $P = 3$ whole-circle RMS discrepancy ranges from 2.82×10^{-4} to 5.85×10^{-2} . The order trend is not pointwise monotone: $P = 2$ beats $P = 3$ on the two alternating tasks.

The depth-three circle campaign also measures nontrivial hidden-feature motion, with relative endpoint motion ranges 0.345441–0.648572, 0.382720–0.736775 and 0.460717–0.677563 in its three hidden layers. Thus the agreement in these instances is not explained by negligible feature motion. Experiments with GELU and sigmoid show that the construction is not tanh-specific, but accuracy is task dependent: the reported GELU hard-outlier $P = 3$ error is 0.02485, whereas harder GELU/sigmoid cases are near 0.26 and 0.24. ReLU and SELU runs are useful empirical probes but fall outside the smooth ODE theorem at their kinks; selected ReLU flows can even be nonunique at exceptional initial states.



(a) Closure versus dense predictions on 1984 held-out images.



Bars: observed refinement margins, not confidence intervals. Open circles: numerical gate failed.

(b) RMS discrepancy across several matched-loss endpoints.

Figure 3: MNIST 3-versus-8, 100 training images, width 4096. At training MSE 0.001, $P = 1, 2, 3$ have held-out RMS discrepancies 0.00324756, 0.00108443, 0.00119922. All declared numerical-refinement gates passed; the slight $P = 3$ worsening shows that autonomous feedback need not make realized error monotone in P .

For this MNIST panel the learned rank bounds are 100, 200, 300. The reported learned-increment storage is 6.25, 12.5 and 18.75 MiB, versus 128 MiB for the unrestricted learned internal correction. The closure also retains the 128 MiB initialized matrix. Including outer blocks, the measured minimum totals are 158.78127, 165.03127 and 171.28128 MiB, versus 152.53125 MiB for dense flow. Integration times were 139.89330, 149.92219 and 152.26930 seconds, versus 102.97555 seconds dense. The empirical achievement is fidelity with a compressed learned-increment representation, not an end-to-end storage or speedup claim.

The frozen experiment record also reports campaign-level compute. The depth-three circle campaign used 2826.286 integration-seconds excluding saved observations (3114.516 including them); the MNIST campaign used 1086.170 and 1115.790 seconds respectively. The shallow moment reproduction used 42 trajectories and 973.36 integration-seconds, while the factor control used 60 trajectories and 2153.56 GPU integration-seconds. The main response-clock panel used 49 fits and 177.89 summed model-seconds; its $P = 4, 5$ extension used 35 fits and 129.00 seconds; the 328-run broad sweep used 37.56 model-minutes over two GPUs. These totals document the scale of the existing evidence rather than a hardware-normalized algorithmic comparison.

8 Limitations and next questions

The main positive result is a fixed-order family of autonomous systems whose state does not grow with elapsed training time and whose physical trajectories converge to dense gradient flow as $P \rightarrow \infty$ on every prescribed finite horizon. Several boundaries matter.

- The initialized random matrices and their transpose actions are retained. Compressing that source while preserving matrix reuse is still open.
- Constants are not uniform in width, depth, sample count or horizon. A direct fixed- P population theorem and a CLT-style finite-width error theory remain future work.
- Neither theorem gives one order valid for all $t \geq 0$. Uniform all-time tracking would require bounded accumulated activity and nonlinear stability with finite total amplification.
- Smooth activations such as tanh, sigmoid, arctangent, GELU, softplus, SiLU, sine and polynomials fall within the stated local smoothness classes. Literal ReLU and standard SELU do not. Their discontinuous selected backward fields require nonsmooth dynamics and a clock that charges jump variation.
- A finite scalar aggregate truncation is a second approximation, beyond history truncation. Differentiating aggregates gives a principled hierarchy, but an economical width-uniform scalar closure is not established here.
- Matching the dense predictor is different from predicting the target well. Generalization requires an additional argument about the function selected by training.

The response-memory construction nevertheless gives a concrete answer to the paper’s question. Exact dynamics can hide a growing history inside a formal population or operator description. Approximate dynamics can replace that history by finitely many evolving coordinates per neuron, retain nonlinear feature feedback, and admit an order-versus-trajectory-error theorem for the complete fixed-width physical path. The key mechanism is specifically neural: paired forward and backward response projections turn the matrix-update defect into a product of two projection errors. Variant A converts that cancellation into compact-horizon $O(P^{-1})$ control, and the jointly

normalized response clock converts it into $O(P^{-2})$ control. The remaining questions concern whether useful orders lie below the explicit moving-state crossover, how the constants scale with width and depth, and whether the retained initialized operators can themselves be represented economically.

The mathematical advance is the representation and trajectory closure itself: it converts coupled endogenous response histories into an autonomous finite-scalar system and controls the complete physical path generated by its own nonlinear feedback. That statement does not rely on a claim of lower total storage, faster execution, empirical verification of the asymptotic rates, or unrestricted historical priority. Those are distinct systems, empirical and historiographic questions; their present limitations bound the scope of the result without weakening the proved cancellation and compact-horizon trajectory theorems.

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